

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

M.Tech. (EC)(C.S.E.) SEM – I, Examination, 2013-14
EC - 704 Digital Signal Processing and Applications

Date: 06/01/2014, Monday

Time: 10.00 a.m to 01.00 p.m

Maximum Marks: 70

Instructions:

1. Attempt all questions.
2. Answer to the two sections must be written in separate answer books.
3. Figures to the right indicate *full* marks.
4. Make suitable assumptions and draw neat figures wherever required.

SECTION-I

- Q-1 a) What is Linear Time Invariant (LTI) system? Give any two examples of time varying systems. [02]
- b) Prove: Frequency Shifting property of Fourier Transform. [03]
- c) Describe: Significance of Region of Convergence (ROC) & its properties. [03]
- d) What is Linear Constant Coefficient Difference Equation (LCCDE)? Prepare mathematical model for second order system. [03]
- Q-2 a) To compute the square root of a positive number A , the iterative algorithm given as: [08]

$$s_n = \frac{1}{2} \left(s_{n-1} + \frac{A}{s_{n-1}} \right), \quad n = 0, 1, \dots$$

Realize a given algorithm using adder, multiplier & delay elements.

- b) Determine the pole-zero plot of the following signal, [04]

$$x(n) = \begin{cases} a^n, & 0 \leq n \leq M-1 \\ 0, & \text{elsewhere} \end{cases}$$

Using z-transform, where $a > 0$ & $M = 8$. Comment on ROC.

OR

- Q-2 a) A discrete time signal $x(n]$ is defined as [04]

$$x(n) = \begin{cases} 1 + \frac{n}{3}, & -3 \leq n \leq -1 \\ 1, & 0 \leq n \leq 3 \\ 0, & \text{elsewhere} \end{cases}$$

- (i) Sketch the signal $x(-n+4)$.
 - (ii) Can you express the signal $x(n)$ in terms of signals $\delta(n)$ and $u(n)$?
- b) Determine Frequency response, Magnitude response and Phase response of the following system, [04]

$$y(n) + \frac{1}{2}y(n-1) = x(n) - x(n-1)$$

- c) Determine z-transform of the following systems: [04]

(i) $y(n) = a^n u(n)$

(ii) $y(n) = x(n) + e^n y(n-1)$

Comment on stability.

- Q-3 a) A LTI system is described by linear difference equation [08]

$$y(n) = 0.2x(n) - 0.5x(n-2) + 0.4x(n-3),$$

For given digital input sequence $\{-1, 1, 0, -1\}$, determine the corresponding digital output sequence.

- b) Determine inverse z-transform of the following system function using partial fraction method, [04]

$$X(z) = \frac{2 + 3z^{-1}}{(1 + z^{-1})(1 + \frac{1}{2}z^{-1})(1 + \frac{1}{4}z^{-1})}$$

Prepare the sequence for $0 \leq n \leq 3$.

OR

- Q-3 a) The input to a causal system Linear Time Invariant (LTI) system is [06]

$$x(n) = -\frac{1}{3}\left(\frac{1}{2}\right)^n u(n) - \frac{4}{3}2^n u(-n-1),$$

and the z-transform of the output is

$$Y(z) = \frac{1 + z^{-1}}{(1 - z^{-1})\left(1 + \frac{1}{2}z^{-1}\right)(1 - 2z^{-1})}$$

(i) Determine impulse response, $h[n]$.

(ii) What is the region of convergence of $Y(z)$?

(iii) Is the system $h(n)$ stable?

- b) Prove properties of z-transform. (*Any three*) [06]

SECTION-II

- Q-4 a) Draw structure of linear phase FIR systems? [02]

- b) What are the Chebyshev low pass characteristics to realize IIR filter? [03]

- c) What is digital image processing? What are the biomedical applications of image processing? [03]

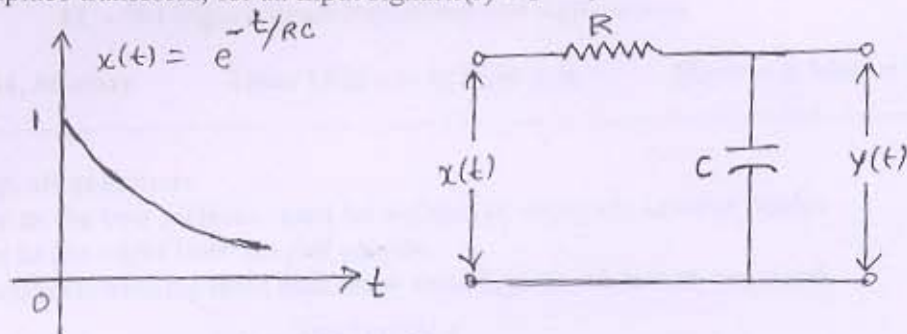
- d) Write short note on: Applications of digital signal processing in telecommunication. [03]

- Q-5 a) Draw the cascade and parallel form implementations of the LTI system with system function [06]

$$H(z) = \frac{(1 + z^{-1})^3}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{8}z^{-1})}$$

- b) Determine the output response of the given low pass RC network using [06]

Laplace transform, for an input signal $x(t) = e^{-t/RC}$.



OR

- Q-5 a) Design a digital Chebyshev filter to satisfy the following constraint [06]
using Bilinear transformation. Assume $T = 1$ sec.

$$0.707 \leq |H(e^{j\omega})| \leq 1 \quad 0 \leq \omega \leq 0.2\pi$$

$$|H(e^{j\omega})| \leq 0.1 \quad 0.5\pi \leq \omega \leq \pi$$

- b) Design an FIR digital filter using rectangular window to approximate [06]
an ideal low pass filter with passband gain of unity, cut-off frequency
of 850 Hz and working at a sampling frequency of $f_s = 5000$ Hz. The
length of the impulse response should be 5.

Q-6 *Attempt Any Two*

- a) An input sequence $x(n) = \{1, 2, 4\}$ is applied to a DSP system having [06]
an impulse response $h[n] = \{1, 2\}$.
i) Determine the Circular Convolution of two sequences with the
padding of zeros using 4-point DFT and IDFT. Verify it with the
graphical method.
ii) Determine the Linear Convolution of two sequences using Circular
Convolution.

- b) Draw the butterfly diagrams using composite-radix DIT FFT, for $N =$ [06]
12. ($N = 4 \times 3$ or $N = 3 \times 4$).
c) Determine 8-point DFT of input sequence $x(n) = \{1, 2, 4\}$, using matrix [06]
method.

OR

- Q-6 a) Compute the N -point DFT of the sequence $x[n] = (-1)^n$, using [06]
decimation in frequency FFT algorithm. Draw neat butterfly diagram
for $N=16$.
b) Write a definition of Discrete Fourier Transform, [06]
Determine total number of:
(i) Trigonometric functions
(ii) Complex & Real multiplications
(iii) Complex & Real additions.